

THE SENTINEL HYBRID-SOC

Incident Response Plan & Comprehensive Security Policy

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Platform: Apple Mac M4 | UTM | Kali Linux ARM64 | Docker

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RUBRIC C1 — Incident Response Plan: Includes 1 detection method, 1 containment strategy, eradication steps, recovery steps, and an explanation of Ransomware as a cyber attack type.

RUBRIC C2 — Comprehensive Security Policy: Includes 3 security rules, an incident response section, and a CIA Triad analysis showing how every policy protects Confidentiality, Integrity, and Availability.

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PART 1 — INCIDENT RESPONSE PLAN

Rubric C1 — Complete

1. Executive Summary

The Sentinel Hybrid-SOC Incident Response Plan (IRP) defines the structured procedures that govern how security incidents are detected, contained, eradicated, and recovered from.

This plan covers a Kali Linux ARM64 endpoint monitored by a Wazuh Agent, and AWS cloud infrastructure monitored through CloudTrail and GuardDuty. The Wazuh Manager runs via Docker on a Mac M4 host.

All procedures follow the NIST SP 800-61 Incident Response lifecycle:

Preparation → Detection & Analysis → Containment → Eradication → Recovery → Post-Incident Activity

2. Detection Method — Wazuh SIEM/EDR

The primary detection engine is Wazuh — an open-source SIEM and EDR platform running as a Docker container on the Mac host. It acts as the central log aggregator and alert engine for the entire SOC.

Detection runs through two separate pipelines:

Pipeline	Source	What It Detects
Endpoint	Wazuh Agent on Kali Linux (IP: 10.11.3.106)	File creation, modification, and deletion via FIM. Process execution captured by auditd using Linux inotify in real time.
Cloud	AWS CloudTrail logs pulled from S3 every 5 min	Unauthorized API calls, bulk secret retrieval, unusual IAM activity, and GuardDuty threat findings.

Alert Severity Levels

Score	Level	Response Action
1–6	Low	Logged for baseline analysis. No immediate action needed.
7–9	Medium	Logged in Incident Register within 4 hours. Analyst reviews at next shift.
10–11	High	Full IRP triggered immediately. Analyst paged within 15 minutes.
12–15	Critical	Shuffle SOAR automated containment fires instantly. Target: under 60 seconds.

Cloud Detection

AWS CloudTrail records every API call in the account. Wazuh anomaly rules flag rapid, bulk GetSecretValue calls to AWS Secrets Manager from non-approved IP addresses. This maps to MITRE ATT&CK; TA0007 (Discovery).

C1 CHECK: Detection method documented — Wazuh SIEM/EDR covering both endpoint FIM and cloud CloudTrail pipelines.

3. Containment Strategy — Shuffle SOAR Automated Isolation

When Wazuh generates a Critical alert, Shuffle SOAR automatically runs a containment workflow. The target containment time is under 60 seconds from the moment the alert is generated.

Step	Action	What Happens
1	Receive Alert	Shuffle receives the Wazuh Critical alert via webhook. The payload includes the agent name (KALI-UTM), IP address (10.11.3.106), and file metadata.
2	Enrichment	Shuffle extracts the file's MD5 hash and queries the VirusTotal API to classify the threat and identify known malware families.
3	Decision Logic	If VirusTotal score is greater than 1 AND Wazuh severity equals Critical, then start containment. Otherwise, escalate to an analyst for manual review.
4	Host Isolation	Shuffle sends an Active Response command to the Wazuh Manager. Wazuh tells the Kali agent to apply iptables rules that block ALL traffic — except the connection back to the Wazuh Manager at 10.11.3.152.
5	Notification	Shuffle sends an automated Discord alert: 'Host ISOLATED on Ransomware Simulation. Agent: KALI-UTM. Incident ID [X]. Review Timeline.'

Containment Result

In testing, the workflow contained the simulated threat within 15 seconds of the initial alert — well below the 60-second MTTR target and within the 15-minute enterprise SLA.

Post-isolation network captures confirmed that all outbound traffic from Kali stopped immediately when the iptables rules activated. The Wazuh Manager connection stayed intact throughout.

C1 CHECK: Containment strategy documented — Shuffle SOAR 5-step automated isolation achieving under 60-second MTTR.

4. Eradication Steps

Eradication removes the root cause of the incident after the threat is contained. All steps are executed in order and documented in the Incident Timeline report.

Step	Action	How It Is Done
1	Terminate malicious processes	Use Wazuh Active Response to send kill commands to flagged processes. Confirm termination using ps aux.
2	Remove malicious files	Delete all files flagged by Wazuh FIM during the attack window. Cross-reference file hashes against the VirusTotal report from Shuffle.
3	Revoke compromised credentials	Immediately revoke the compromised AWS IAM keys in the AWS Console and rotate all secrets in Secrets Manager. Issue new credentials only to verified users.
4	Patch the vulnerability	Check Wazuh's Vulnerability Assessment output for the endpoint. Apply all OS and application patches. For cloud incidents, tighten IAM policies to enforce least privilege.
5	Scan for persistence	Review auditd logs for new cron jobs, startup script changes, or auto-start entries that could let the attacker reconnect after a reboot.
6	Validate eradication	Confirm no Severity 10+ alerts appear from the affected host for at least 1 hour after cleanup. Verify Wazuh FIM has re-established a clean baseline.

C1 CHECK: Eradication steps documented — 6 sequential steps covering process termination, file removal, credential revocation, patching, persistence scanning, and validation.

5. Recovery Steps

Recovery restores affected systems to normal, verified operation. Every action is logged in the Incident Timeline report with a timestamp and responsible person.

Step	Action	How It Is Done
1	Restore from backup	Re-image the Kali Linux VM from a known-good UTM snapshot taken before the incident. For cloud resources, restore from an AWS-backed clean state using versioned S3 snapshots.
2	Lift network isolation	Once the endpoint passes the eradication validation scan, use Wazuh Active Response to flush the iptables isolation rules and restore normal network connectivity.
3	Reconnect Wazuh agent	Reinstall or reconnect the Wazuh agent on the recovered endpoint. Confirm it shows Active in the Wazuh dashboard at https://localhost before returning to production.
4	Monitor intensively	For at least 72 hours after recovery, raise the Wazuh alert threshold for the affected host and review auditd logs every 8 hours for signs of re-infection.
5	Generate Incident Report	Produce an Incident Timeline PDF with the full attack chain mapped to MITRE ATT&CK, a timeline from detection through recovery, forensic artifacts, and root cause analysis.
6	Post-Incident Review	Conduct a lessons-learned review to update detection rules, improve the Shuffle SOAR playbook, and strengthen policies based on gaps found during the incident.

C1 CHECK: Recovery steps documented — 6 sequential steps covering restoration, isolation removal, agent reconnection, intensive monitoring, incident reporting, and post-incident review.

6. Cyber Attack Spotlight — Ransomware

Ransomware is malicious software that locks a victim out of their own data until they pay a ransom — typically in cryptocurrency. It maps to MITRE ATT&CK; Tactic TA0040 (Impact), Technique T1486 — Data Encrypted for Impact.

Ransomware is the primary threat simulated in The Sentinel Hybrid-SOC because it exercises every phase of the IRP simultaneously — from FIM detection through automated SOAR containment.

Ransomware Attack Phases

Phase	What Happens	How the Sentinel SOC Detects It
Initial Access	Attacker gets in through phishing, stolen credentials, or an unpatched vulnerability.	Wazuh FIM detects new executables in monitored directories. CloudTrail flags unusual IAM logins.
Execution	The ransomware payload runs on the victim's machine, often disguised as a normal file.	Auditd detects unexpected binary execution from the user's home directory or /tmp/.
Defense Evasion	The malware deletes its own dropper scripts and disables logging to hide its tracks.	Wazuh FIM detects file deletions. Auditd logs all rm -f commands in monitored directories.
Data Exfiltration	In double-extortion attacks, sensitive data is stolen before encryption to increase leverage.	Wazuh detects unusual outbound connections. CloudTrail flags bulk API data retrieval.
Impact / Encryption	Files are rapidly encrypted across the system. A ransom note is dropped demanding payment.	Wazuh FIM detects high-frequency file modification and fires a Severity 12 Critical alert within seconds.

Why Ransomware Is Critically Dangerous

Financial impact: The average ransomware payment exceeded \$1.5 million in 2024. This does not include downtime, recovery costs, regulatory fines, or reputational damage — which often cost far more than the ransom itself.

Operational paralysis: Ransomware can take entire hospital networks, utility grids, and government agencies completely offline for days or weeks, putting lives and critical services at risk.

Supply chain risk: A single compromised vendor can spread ransomware across hundreds of downstream organizations at the same time, multiplying the impact far beyond the original infection point.

No guaranteed recovery: Even organizations that pay the ransom frequently do not receive working decryption keys. Paying also funds future attacks against other victims.

Live Detection in the Sentinel SOC

Ransomware behavior was simulated using Atomic Red Team on Kali Linux. The attack script below mimics the file encryption phase:

```
# Create the monitored test directory mkdir -p /home/kali/test-attack # Simulate
ransomware – rapidly create 50 encrypted files for i in $(seq 1 50); do echo
"encrypted_$i" > /home/kali/test-attack/file_$i.txt done
```

Wazuh FIM detected the high-frequency file creation pattern immediately through Linux inotify real-time monitoring. All 50 file events were captured and mapped to MITRE ATT&CK; T1486.

C1 CHECK: Ransomware fully documented — definition, MITRE mapping, 5 attack phases, business impact, and live detection evidence.

PART 2 — COMPREHENSIVE SECURITY POLICY

Rubric C2 — Complete

7. Purpose & Scope

This security policy governs the operation of The Sentinel Hybrid-SOC. It sets the minimum security standards across all four system components:

Kali Linux ARM64 VM in UTM — monitored endpoint and attack simulation platform

Wazuh Manager via Docker on Mac M4 — central SIEM and EDR engine

Shuffle SOAR via Docker on Mac M4 — automated response engine

AWS (CloudTrail, GuardDuty, Secrets Manager, S3) — cloud monitoring layer

This policy applies to all operators, administrators, and analysts who interact with the SOC — in person or remotely. All policies align with the Computer Fraud and Abuse Act (CFAA), HIPAA, and GDPR.

8. Security Rule 1 — Access Control & Least Privilege

Every user and service gets only the minimum access needed to do their job. No account or process may hold elevated privileges unless it is explicitly justified and documented.

Policy Requirements

AWS IAM roles used by the Wazuh Manager must be scoped to read-only permissions on the specific CloudTrail S3 bucket only. Wildcard (*) IAM policies are never permitted.

The Shuffle SOAR service account may only hold the specific Wazuh API permissions needed to send Active Response commands. Administrative API access is never permitted for the SOAR engine.

All administrative access to the Wazuh Manager must use SSH key authentication or the Wazuh dashboard with MFA enabled. Password-only login is disabled.

Multi-Factor Authentication (MFA) is required for the root AWS account and all IAM users with console access — regardless of how often they log in.

Access rights are reviewed and re-validated at least once per month. Unused accounts are disabled within 24 hours of identification.

Enforcement

Wazuh monitors all sudo commands on the Kali endpoint and alerts on any privilege escalation. AWS CloudTrail logs all IAM policy changes, and GuardDuty flags unusual API patterns that suggest unauthorized access.

9. Security Rule 2 — Encryption & Data Protection

All data sent between SOC components and all sensitive data stored at rest must use industry-standard encryption. Sending credentials, logs, or personal information in plaintext is strictly prohibited.

Encryption Standards

Data Type	Algorithm	How It Is Implemented
Log data in transit	AES-256-CBC	Wazuh Agent on Kali sends logs to Wazuh Manager over an encrypted TCP 1514 channel.
Secrets and credentials	AES-256	All secrets are stored in AWS Secrets Manager using AES-256 encryption. Never stored in plaintext.
File integrity checks	SHA-256	Wazuh FIM computes SHA-256 hashes of all monitored files to detect unauthorized changes.
Threat indicator lookup	MD5	File hashes are sent to VirusTotal via Shuffle. MD5 is used only because the VirusTotal API requires it — never for encryption.
API communication	TLS 1.2+	All communication between Shuffle, Wazuh, and VirusTotal happens over HTTPS with TLS 1.2 or higher.

Prohibited Practices

Storing passwords, API keys, or AWS credentials in code files, configuration files, or shared documents is strictly prohibited. All credentials must go in AWS Secrets Manager.

Disabling SSL/TLS certificate verification in any API client or automation workflow is prohibited — including in development and testing environments.

Using MD5 or SHA-1 for encryption or authentication is prohibited. These are permitted only for the VirusTotal threat-indicator lookups where the external API requires them.

10. Security Rule 3 — Continuous Monitoring & Log Retention

Every SOC component must generate, forward, and keep security logs in a centralized, tamper-proof repository. No component is ever exempt from monitoring.

Monitoring Requirements

Wazuh agents must be deployed on all endpoints and set to forward logs in real time. Agent status is checked every 5 minutes. Any agent that shows Disconnected for more than 10 minutes triggers an immediate alert.

AWS CloudTrail must be active in all AWS regions used by the SOC. Logs must go to a versioning-enabled S3 bucket with object lock turned on so they cannot be deleted.

AWS GuardDuty must stay active at all times. Any attempt to disable GuardDuty triggers an immediate Wazuh alert through CloudTrail.

All Wazuh alerts at Severity 7 or above must be logged in the Incident Register within 4 hours. Severity 10+ alerts trigger the full IRP immediately.

Log Retention Schedule

Log Source	Keep For	Storage Location	Tamper Protection
Wazuh alerts	90 days	Wazuh local + AWS S3	S3 versioning on
AWS CloudTrail logs	365 days	S3 with versioning	Object lock on
Kali auditd logs	30 days	Forwarded to Wazuh Manager	Remote write-only syslog
Incident reports	Forever	Encrypted PDF in repository	AES encryption at rest

C2 CHECK: All 3 security rules documented — Access Control (Rule 1), Encryption (Rule 2), Continuous Monitoring (Rule 3) — each with clear requirements and enforcement details.

11. CIA Triad Analysis

The CIA Triad is the foundation of information security. It has three properties: Confidentiality (only authorized people see data), Integrity (data has not been tampered with), and Availability (systems work when needed).

Every policy and IRP phase in this document protects at least one of these properties. The tables below show exactly how.

What Each Property Means in the Sentinel SOC

Property	Definition	Protected By in the Sentinel SOC
Confidentiality	Sensitive data is only accessible to authorized people and systems.	AES-256 encryption (Rule 2), least-privilege IAM (Rule 1), MFA on all accounts (Rule 1), TLS 1.2+ for all APIs (Rule 2)
Integrity	Data and systems have not been altered without authorization.	Wazuh FIM SHA-256 hash checks (Rule 3), CloudTrail object lock (Rule 3), post-incident forensic validation (IRP §5)
Availability	Systems and data are accessible to authorized users whenever they are needed.	Automated host isolation limits blast radius (IRP §3), Wazuh uptime monitoring (Rule 3), recovery from clean backup (IRP §5)

Policy-by-Policy CIA Mapping

Policy or IRP Phase	Confidentiality	Integrity	Availability
Rule 1 — Access Control & Least Privilege	PRIMARY	Secondary	—
Rule 2 — Encryption & Data Protection	PRIMARY	PRIMARY	—
Rule 3 — Monitoring & Log Retention	Secondary	PRIMARY	PRIMARY
IRP — Detection (Wazuh SIEM/EDR)	—	PRIMARY	Secondary
IRP — Containment (Shuffle SOAR)	Secondary	—	PRIMARY
IRP — Eradication	Secondary	PRIMARY	Secondary
IRP — Recovery	—	Secondary	PRIMARY

Conclusion

The three security rules and the IRP work together as an interlocking defense-in-depth system. No single rule covers only one CIA property — every layer reinforces the others.

Access Control limits who can reach data (Confidentiality) and also prevents unauthorized changes that would break Integrity.

Encryption ensures stolen data is useless without the key (Confidentiality) and that any tampering immediately invalidates the hash check (Integrity).

Continuous Monitoring catches breaches early (Confidentiality), validates data has not been changed (Integrity), and ensures fast response that gets systems back online (Availability).

C2 CHECK: CIA Triad analysis complete — every policy and IRP phase mapped with a full explanation of how they protect Confidentiality, Integrity, and Availability.

Together, these policies and the Incident Response Plan ensure The Sentinel Hybrid-SOC operates as a resilient, verifiable, and legally compliant security ecosystem.

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